Worksheet W6 · W-Seminar "Time"

Design your study

Before you collect a single number: find the kind of data that can really answer your question – and talk your plan through.

1 My question

from W4 – research question and working thesis / hypothesis

Question: How does the tempo of background music change how long a wait feels?

Thesis: Students who wait with fast music estimate a 3-minute wait as shorter than with slow music.

2 Which quality criterion is at risk?

Objectivity: same result whoever runs or scores it · Reliability: same result again, little noise · Validity: measures what you claim. Tick the criterion that is hit first.

	O	R	V
a) You mark students' open answers yourself – without a marking key.	☒	☐	☐
b) One single question ("Are you stressed?") is your whole measure of stress.	☐	☒	☐
c) A speed-reading task is used to show how well students understand a novel.	☐	☐	☒
d) Group A waits before lunch, group B after lunch.	☐	☐	☒

3 Five ways to get data – brainstorm for *your* question

Think 4 min Fill in what each type could give you – even the odd ones; a weird idea often leads to a good one. Pair 4 min Swap sheets. Your partner ticks the option that convinces them most and writes one question in the margin.

type	what exactly could I collect?	+ / –	best?
Survey what people say	ask 100 students how long typical waits feel – with and without music	+ many people; – memory, vague answers	☐
Test what people can do	(hardly fits: perception is not a skill)	– wrong construct	☐
Experiment what causes what	3-minute wait with fast / slow music, estimate in minutes	+ shows cause; – rooms, time, permission	☒
Coding texts count categories	how song lyrics or novels describe waiting	+ no participants; – a different question	☐
Existing data others' statistics	results of published music-and-waiting studies	+ large samples; – not my own data	☐

My partner's question: Would people notice the music at all if they are on their phones?

4 Talk it through: partner interview

Pair 2 × 6 min Your partner asks the questions and notes your answers in keywords. Then swap roles. Don't defend – explain; "I don't know yet" is a fine answer and goes on your to-do list.

partner asks	answers (keywords)
1 What exactly do you measure – and how does it become a number or category?	estimated waiting time in minutes, written on a card
2 Who takes part, how many, and how are they chosen?	Year 10–11, 2 × 30, groups drawn by lot
3 What else could explain your result?	room, time of day, knowing the song, phones
4 Which result would show that your thesis is wrong?	no difference – or slow music feels shorter
5 What could go wrong in practice – what is your plan B?	no permission → online version at home

My partner's best tip: add a question afterwards: how fast did the music feel?

5 My design

decide – you can still change it after the pilot

☐ survey ☐ test ☒ experiment ☐ coding texts ☐ existing data ☐ mixed: _____

Idea → rule (operationalisation): "time flies" → estimate of a 3-min wait in minutes

Variables / items / categories: IV tempo (fast / slow), DV estimate; same room, volume

Sample (who, n, how chosen): 2 × 30 students, Year 10–11, random assignment

Procedure, step by step:

1. same room and script 2. 3 minutes of music 3. estimate on a card 4. tempo rating

6 Secure the three criteria in *my* study

Share 1 min each Present your design to another pair. They name the biggest risk they see – add it to the table.

	biggest risk	what I do about it
Objectivity	I explain things differently	written script, same for all
Reliability	rough timing	stopwatch, exactly 180 s
Validity	confounds, weak manipulation	same room and time, tempo rating

Before I start ☒ asked my teacher ☐ school permission / parents' consent ☒ anonymous codes ☒ pilot with 5 people